1 Why "random phase"?

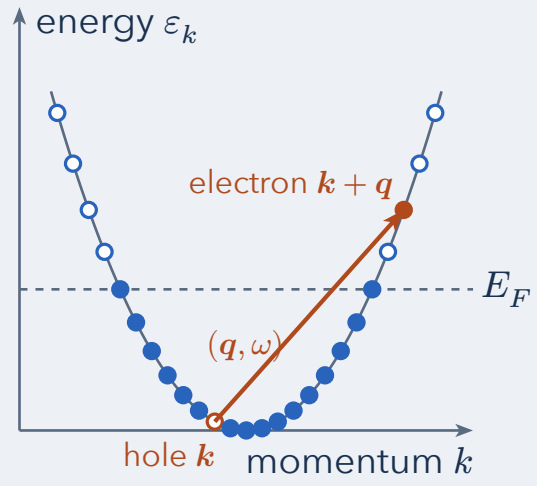
A density wave $\rho_{\boldsymbol{q}} = \sum_j e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_j}$ is driven by products $\rho_{\boldsymbol{q}-\boldsymbol{q}'}\rho_{\boldsymbol{q}'}$. Add up the phases of N electrons:

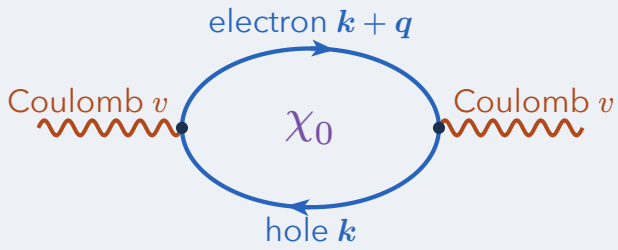


For $N \to \infty$ the $\boldsymbol{q}' \neq \boldsymbol{q}$ terms are negligible ($\sqrt{N} \ll N$) and are dropped. Each Fourier component then responds only to the total (external + induced) potential at the same $\boldsymbol{q}$: the time-dependent Hartree approximation.



2 Independent-particle response χ_0

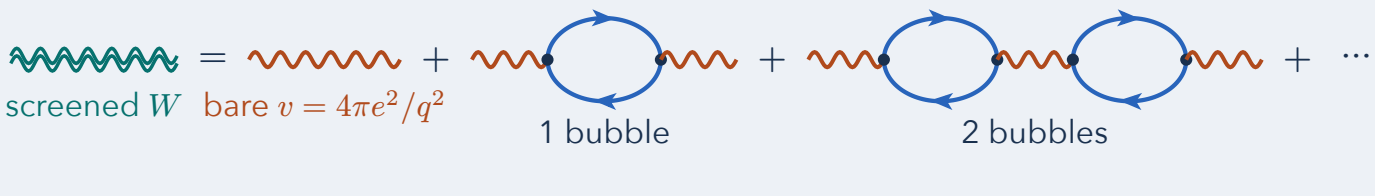




$$\chi_0(\boldsymbol{q},\omega) = \tfrac{2}{V}\sum_{\boldsymbol{k}} \tfrac{f_{\boldsymbol{k}} - f_{\boldsymbol{k}+\boldsymbol{q}}}{\omega + \varepsilon_{\boldsymbol{k}} - \varepsilon_{\boldsymbol{k}+\boldsymbol{q}} + i\eta}$$

One electron-hole pair, created and destroyed by the field. f are occupations, so only pairs that cross the Fermi level E_F contribute. Poles sit at the pair energies $\varepsilon_{\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{\boldsymbol{k}}$. In diagrams $\chi_0 = -iGG$.

3 Summing bubble chains: the RPA screened interaction W



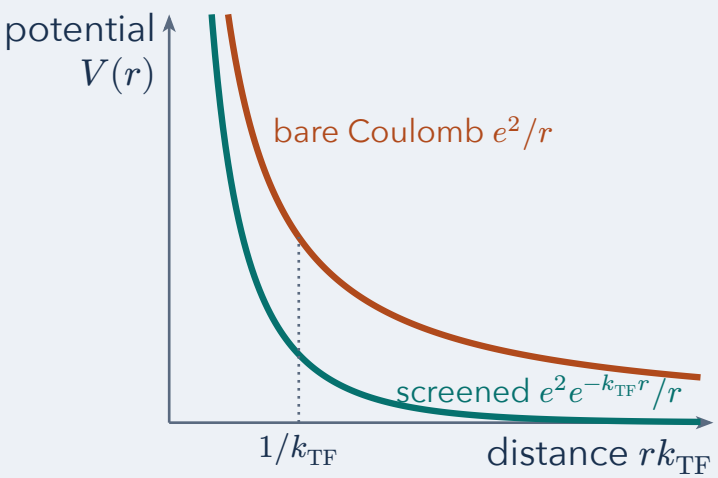
$$W = v + v\chi_0 v + v\chi_0 v\chi_0 v + \ldots$$
$$= v + v\chi_0 W = \tfrac{v}{1 - v\chi_0} = \varepsilon^{-1} v$$

with the RPA dielectric function $\varepsilon = 1 - v\chi_0$

Each bubble is an electron-hole pair polarized by the field of the previous one. RPA sums every chain exactly but drops exchange and ladder (electron-hole attraction) diagrams between bubbles, i.e. vertex corrections.

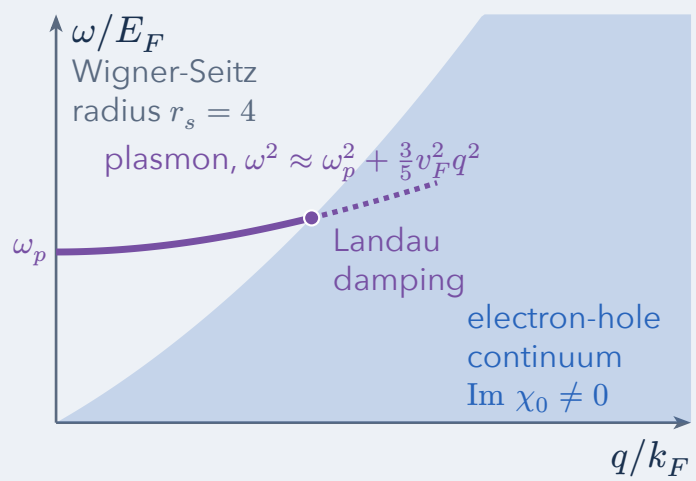
4 Static screening

static dielectric function
$\varepsilon(q,0) \approx 1 + k_{\mathrm{TF}}^2/q^2$ for $q \to 0$
Thomas-Fermi wavevector k_{TF}



5 Plasmons

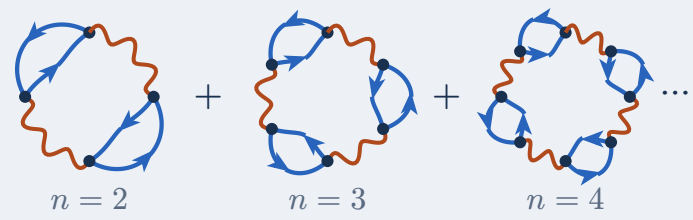
collective mode where $\varepsilon(\boldsymbol{q},\omega) = 0$
plasma frequency ω_p, Fermi velocity v_F
Fermi wavevector k_F, Fermi energy E_F



6 Correlation energy

RPA correlation energy

$$E_c^{\mathrm{RPA}} = \int_0^\infty \tfrac{d\omega}{2\pi}\,\mathrm{Tr}[\ln(1 - \chi_0(i\omega)v) + \chi_0(i\omega)v]$$
$$= -\sum_{n\geq 2} \tfrac{1}{n}\int_0^\infty \tfrac{d\omega}{2\pi}\,\mathrm{Tr}\,(\chi_0(i\omega)v)^n$$



Closed rings of bubbles. Used for total energies via the adiabatic-connection fluctuation-dissipation theorem: captures van der Waals dispersion, but absolute E_c comes out too negative.